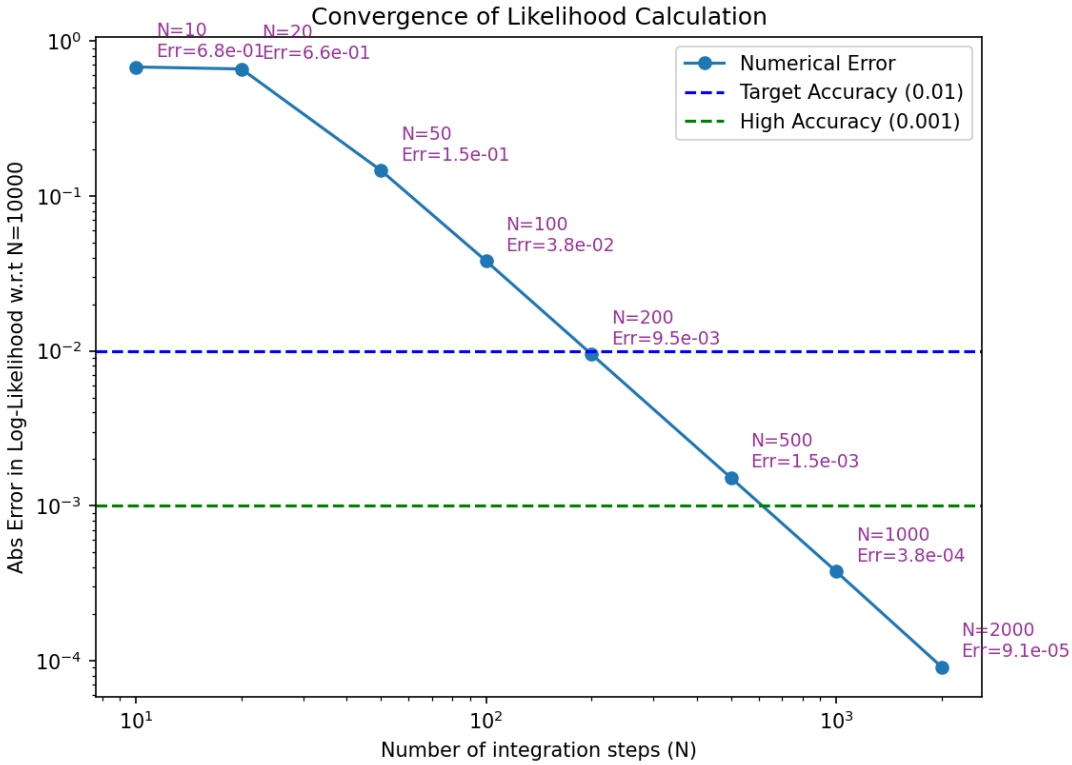
Unit 4: Likelihoods & Optimization Results

Ling Qingyang s2471545

1. Likelihood Convergence Test

To ensure the accuracy of the likelihood calculation, I tested the convergence of the Log-Likelihood value as a function of the number of integration steps N. I compared the results against a high-precision reference calculated with N=10,000.

The convergence test results are plotted below:



Analysis:

The plot shows a clear power-law convergence of the numerical error.

High-precision Reference Log-L (N=10000): -587.68734

N= 10 | Log-L=-587.00812 | Diff=0.67922

N= 20 | Log-L=-587.02768 | Diff=0.65966

N= 50 | Log-L=-587.54101 | Diff=0.14633

N= 100 | Log-L=-587.64935 | Diff=0.03799

N= 200 | Log-L=-587.67783 | Diff=0.00951

N= 500 | Log-L=-587.68582 | Diff=0.00152

N=1000 | Log-L=-587.68697 | Diff=0.00038

N=2000 | Log-L=-587.68725 | Diff=0.00009

Decision:

I chose **N=500** for the subsequent optimization tasks. At this resolution, the numerical error (1.5*10^(-3)), which is negligible to 1. This ensures that the integration error does not affect the parameter estimation while maintaining computational efficiency.

---

2. Optimization of the Standard Model

Using the scipy.optimize.minimize function, I maximized the Gaussian likelihood (by minimizing the negative log-likelihood) to find the best-fitting parameters for the standard ΛCDM model (varying Omega_m, Omega_Λ, and H0).

2.1 Best-Fit Parameters

The optimization converged successfully with the following results:

--- Optimization Results ---

Best fit Omega_m : 0.35030

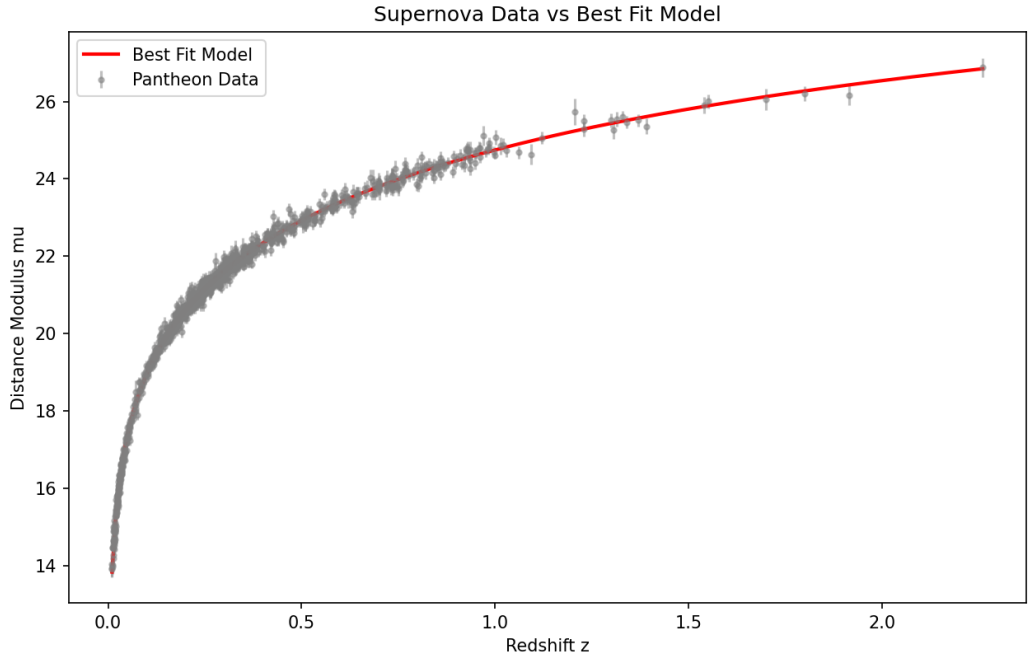
Best fit Omega_lambda : 0.83099

Best fit H0 : 72.15239

Maximum Log-Likelihood: -516.34335

2.2 Model vs. Data Plot

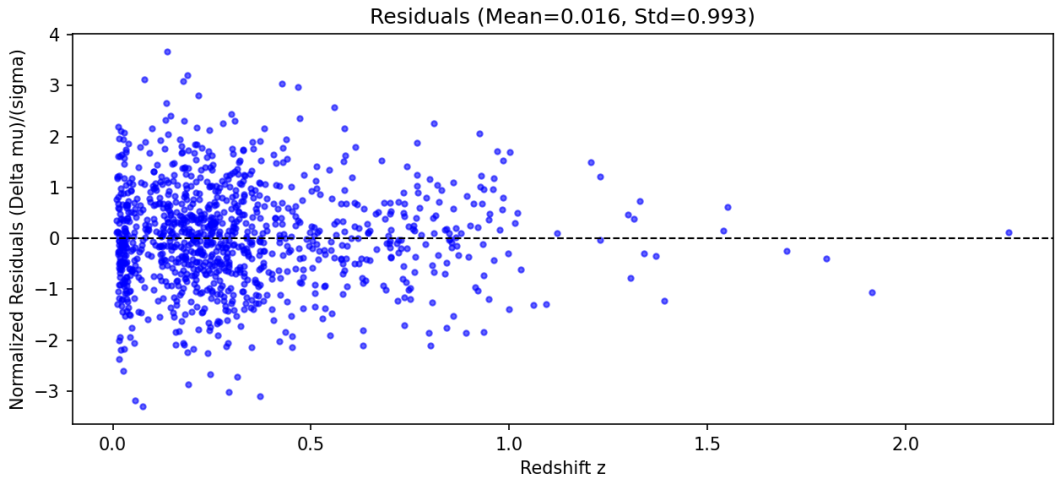
The plot below shows the Pantheon supernova data (with error bars) and the theoretical distance modulus curve calculated using the best-fit parameters derived above.



Visually, the red model curve passes centrally through the data points across the entire redshift range, indicating a good fit.

2.3 Residuals Analysis

To quantitatively assess the quality of the fit, I calculated the normalized residuals:



The statistics of the residuals are:

- **Mean:** 0.01644
- **Standard Deviation:** 0.99253

Comment:

The mean of the residuals is very close to zero, indicating no significant systematic bias in the model. The standard deviation is very close to 1.0, which suggests that the model explains the data well and that the observational errors sigma_i are accurately estimated. The distribution of points in the plot appears random without any obvious structure, further confirming the quality of the fit.

---

3. Changing the Model (Comparison)

I repeated the optimization process for a second model where Dark Energy is assumed to be non-existent (fixing Omega_Lambda = 0).

3.1 No-Dark Energy Results

Best fit Omega_m: 0.00000

Best fit Omega_lambda: 0.00000 (Fixed)

Best fit H0: 69.04068

Maximum Log-Likelihood: -577.31577

3.2 Comparison and Conclusion

--- Model Comparison ---

Log-L (Standard) : -516.34335

Log-L (No Lambda) : -577.31577

Conclusion: Standard Model (with Dark Energy) fits BETTER.

Comment on Parameter Values:

In the model without Dark Energy, the optimizer drove the matter density parameter Omega_m to **0.0** (effectively an Empty Universe). This happens because, without the accelerating expansion provided by Lambda, the model tries to maximize the distance to the supernovae by removing all matter (since matter causes deceleration). However, even an empty universe (which expands at a constant rate) cannot match the faintness of the distant supernovae as well as the accelerating universe of the Standard Model.